

Study Sync Academic Collaboration Platform

Presented By:

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Course: Software Design and Analysis (SDA)
Instructor: Ahmed Qaiser



Overview of Project:

A centralized solution designed to keep students organized and connected. Bringing assignment tracking, class scheduling, and structured project collaboration into one place.



Assignment Tracking



Team Collaboration



Schedule Management



The Problem

Current academic challenges faced by students:

- **Deadlines overlap and students struggle to keep track of multiple assignments across different courses.**
- **Group projects become chaotic with no structured collaboration tools.**
- **Important class material ends up scattered across chats, drives, and email threads.**
- **No single organized space to manage academic tasks and resources.**





The Solution

Study Sync platform features:

- **Track deadlines with priority tags and automated reminders.**
- **Structured project collaboration with shared tasks.**
- **Curated learning materials from instructors.**
- **Real-time updates for deadlines and activities.**





System Actors

5 user roles with different permissions:

1. Student:

Tracks deadlines, schedules, resources

2. Group Member:

Collaborates on projects

3. Senior Students:

Shares resources & guidance

4. Instructor:

Publishes materials & assignments

5. Administrator:

Manages users & moderation



Core Functionalities

9+ features including innovative AI-assisted scheduling:

Authentication:
Secure login with role-based access control.

Assignment Tracking:
Track deadlines with priority tags.

Class Schedule:
Weekly timetable management

Project Collaboration:
Shared tasks, files, discussions

Smart Study Planner:
AI Assistant

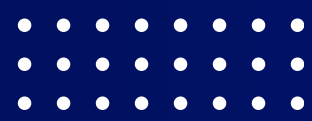
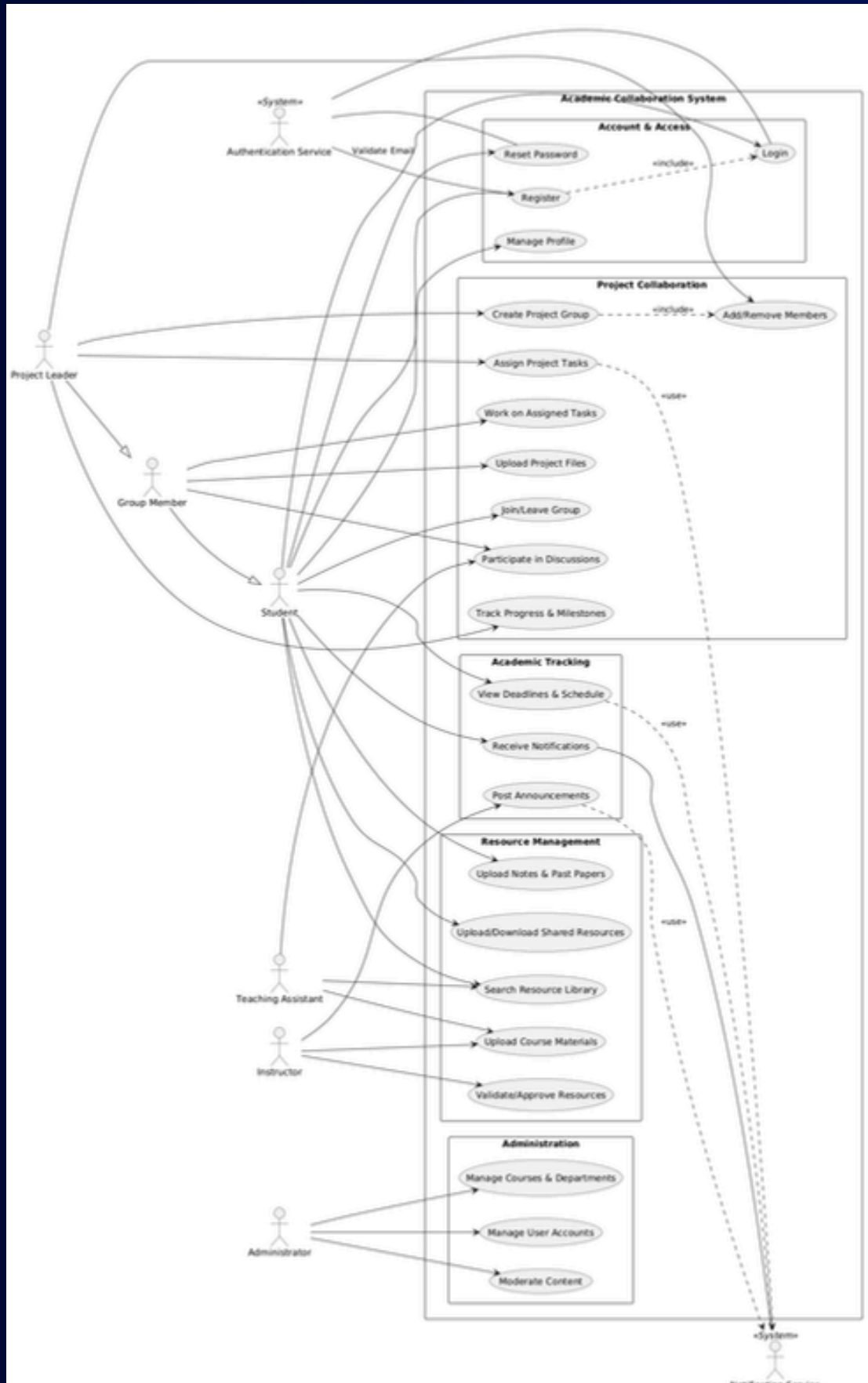
Material Sharing:
Instructor uploads & resources

Search & Filter:
Keyword-based resource search

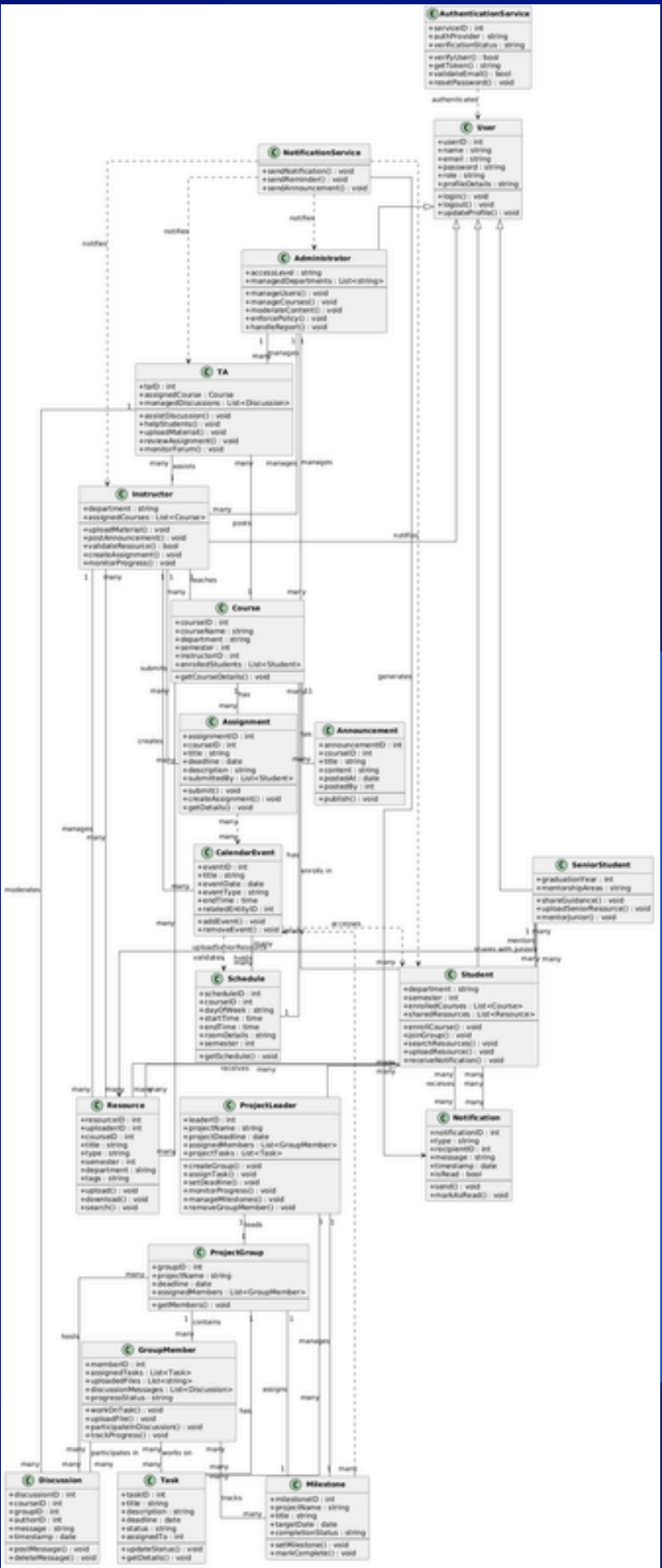
Notifications:
Real-time updates & alerts

Admin Module:
Content review & moderation

Use Case Diagram:



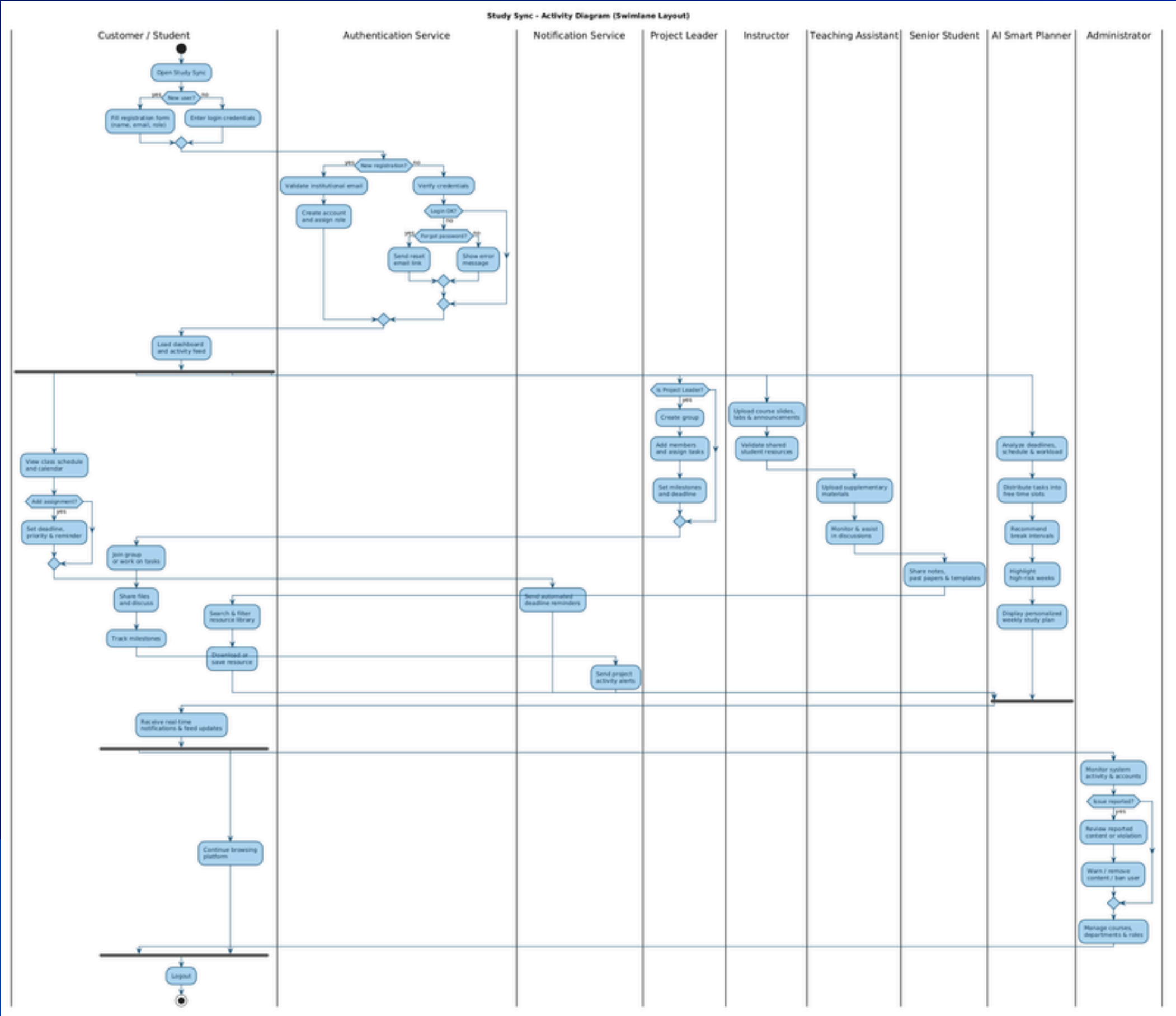
UML Class Diagram:



Sequence Diagram:



Activity Diagram:

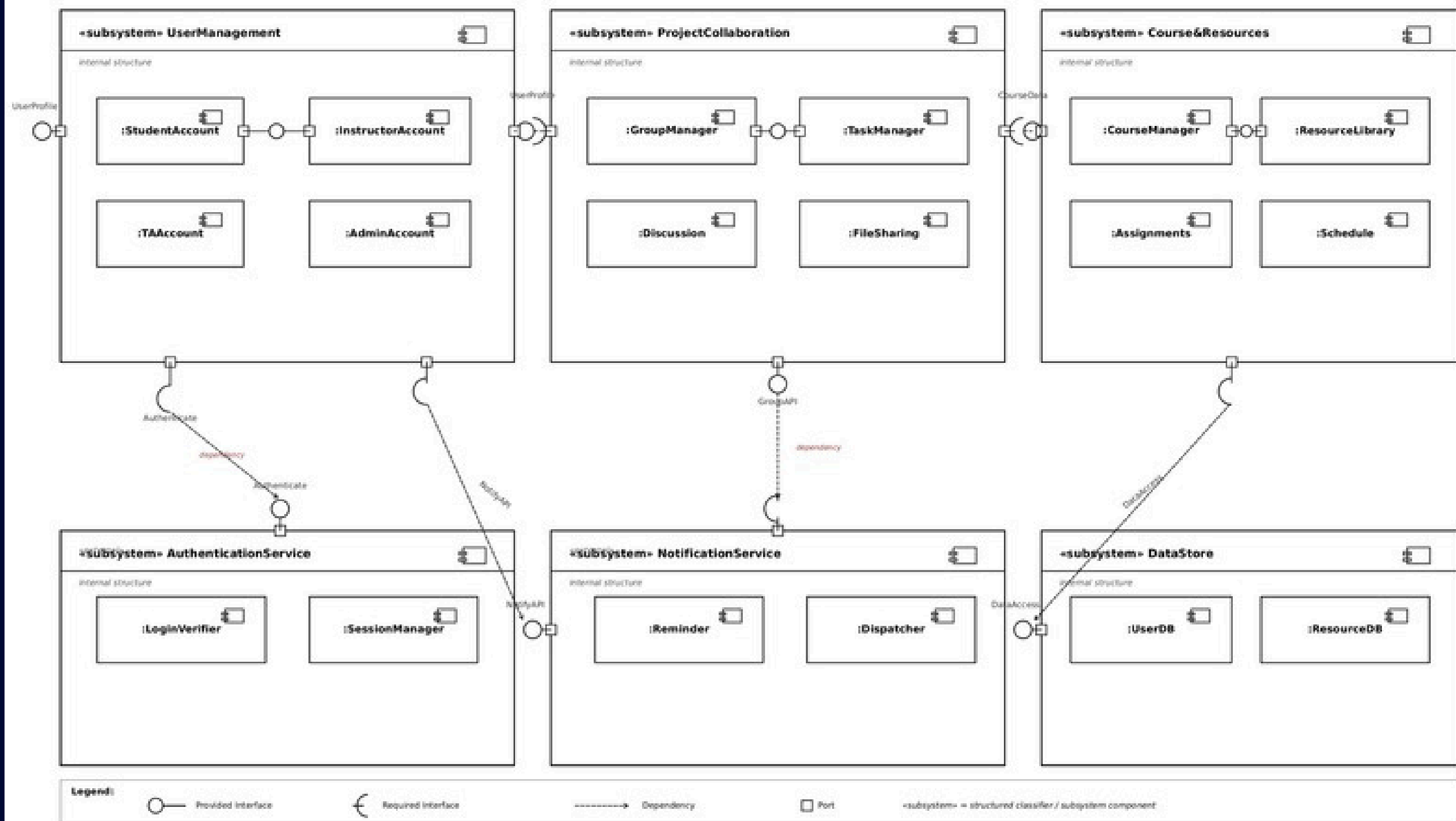


Component Diagram:



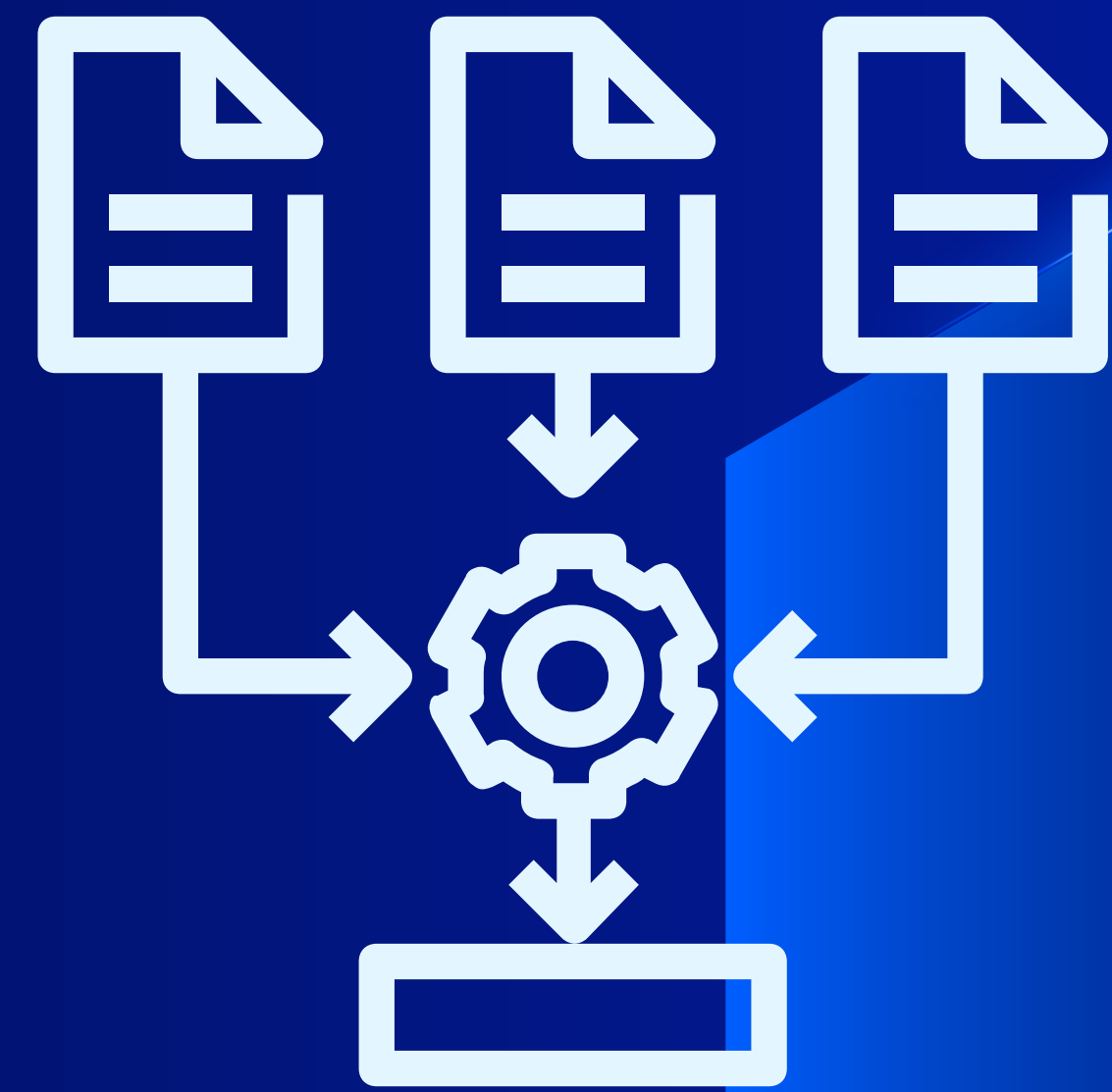
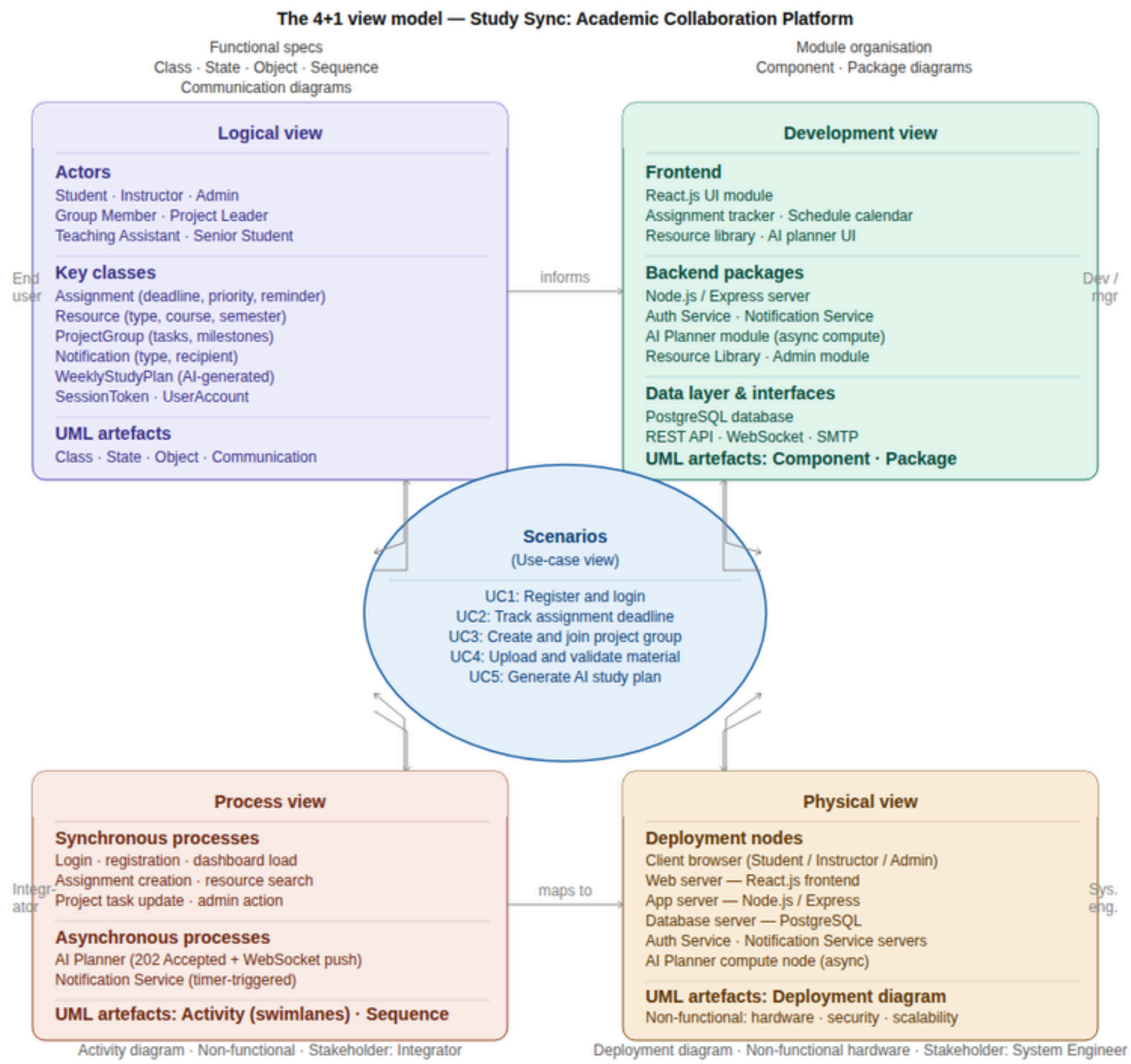
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Study Sync: Academic Collaboration Platform — UML Component Diagram



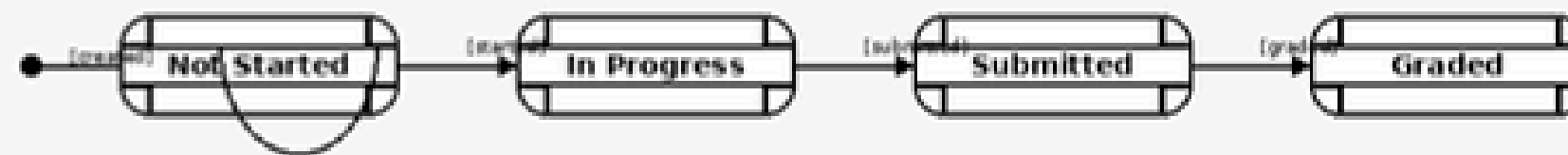
4+1 View Model Diagram:

10



State-Chart Diagrams

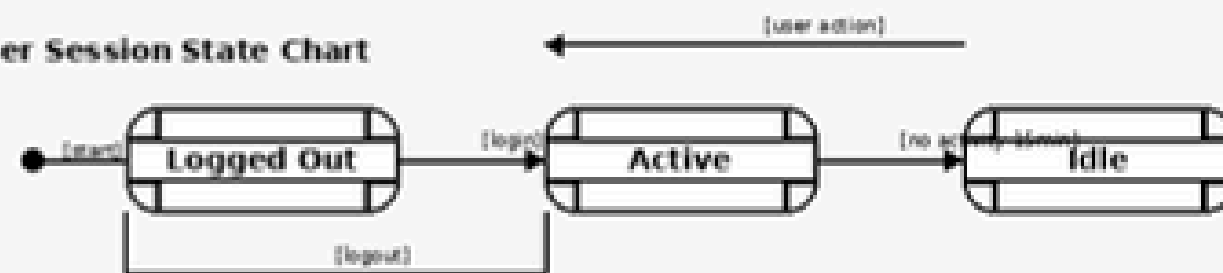
1. Assignment State Chart



2. Project Group State Chart



3. User Session State Chart

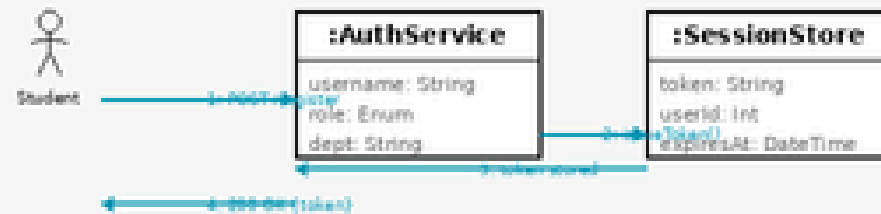


4. Resource Moderation State Chart



Communication Diagram

Scenario 1: User Registration & Login



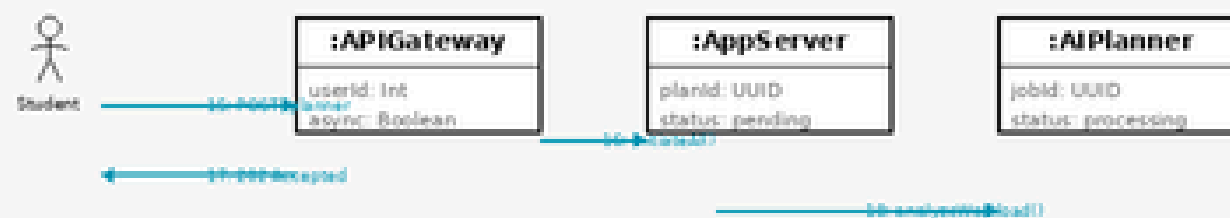
Scenario 2: Dashboard Load & Assignment Tracking



Scenario 3: Project Group & Task Management



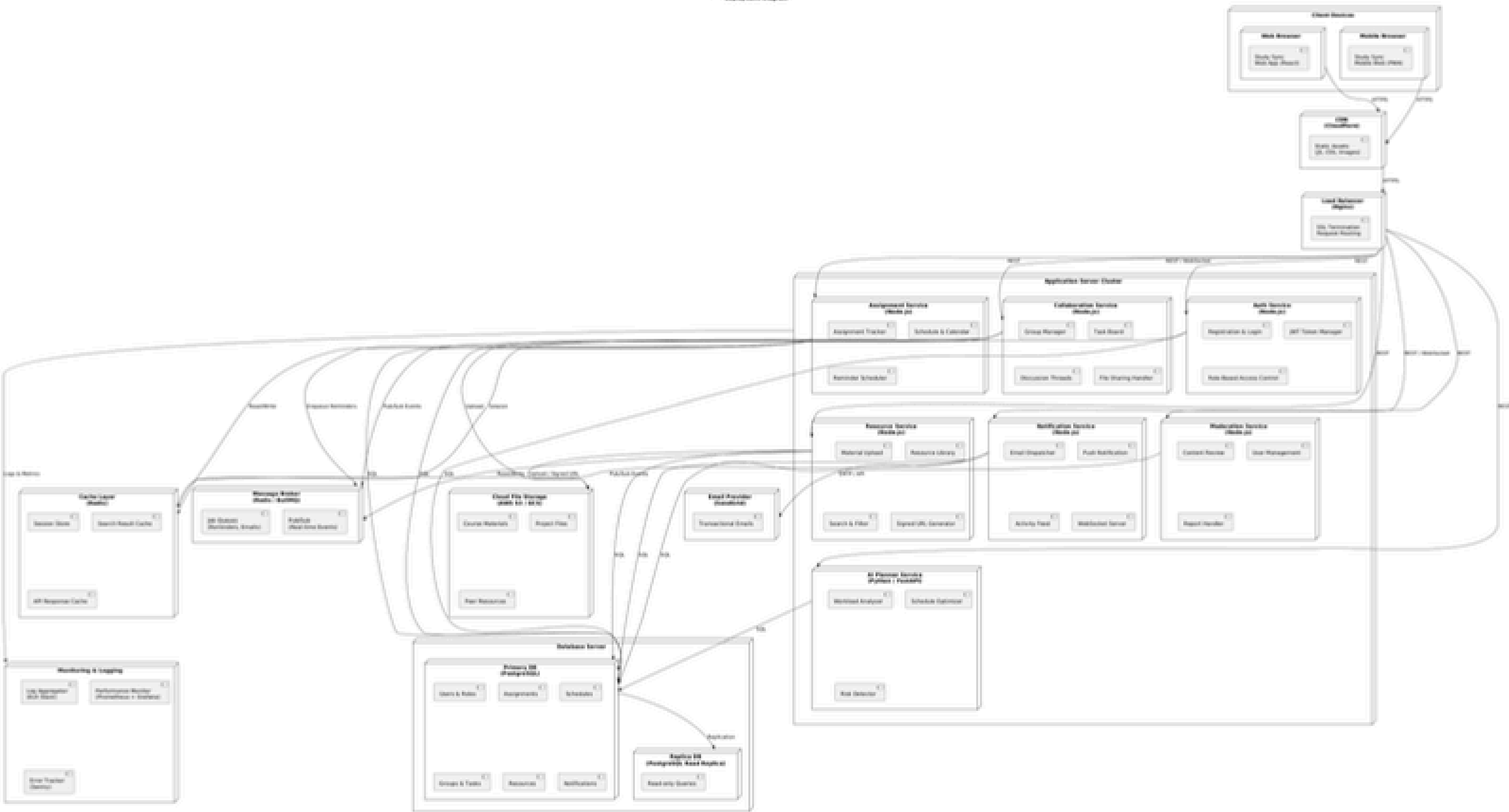
Scenario 4: AI Study Planner (Async, Non-blocking)



Deployment Diagram:



UML Title - Academic Collaboration Platform
Deployment Diagram



Singleton Pattern

Use Case: Notification Manager

```
#include <iostream>
using namespace std;
```

```
class NotificationManager {
private:
    static NotificationManager* instance;

    // Private constructor
    NotificationManager() {}

public:
    // Get instance
    static NotificationManager* getInstance() {
        if (instance == nullptr) {
            instance = new NotificationManager();
        }
        return instance;
    }
};
```

```
void sendNotification(string message) {
    cout << "Notification: " << message << endl;
}
};
```

```
// Initialize static variable
NotificationManager* NotificationManager::instance =
nullptr;
```

```
// Usage
int main() {
    NotificationManager* n1 =
NotificationManager::getInstance();
    NotificationManager* n2 =
NotificationManager::getInstance();

    n1->sendNotification("Assignment due tomorrow");

    cout << (n1 == n2) << endl; // 1 (true)
}
```

Factory Pattern

Use Case: Creating Users (Student, Instructor, Admin)

```
#include <iostream>
using namespace std;
```

```
// Base class
class User {
public:
    virtual void showRole() = 0;
};
```

```
// Derived classes
class Student : public User {
public:
    void showRole() {
        cout << "I am a Student" << endl;
    }
};
```

```
class Instructor : public User {
public:
    void showRole() {
        cout << "I am an Instructor" << endl;
    }
};
```

```
class Admin : public User {
public:
    void showRole() {
        cout << "I am an Admin" << endl;
    }
};
```

```
// Factory
class UserFactory {
public:
    static User* createUser(string type) {
        if (type == "student")
            return new Student();
        else if (type == "instructor")
            return new Instructor();
        else if (type == "admin")
            return new Admin();
        return nullptr;
    }
};
```

```
// Usage
int main() {
    User* u1 = UserFactory::createUser("student");
    User* u2 = UserFactory::createUser("admin");

    u1->showRole();
    u2->showRole();
}
```

Adapter Pattern

Use Case: Integrating External Calendar API

```
#include <iostream>
using namespace std;

// Target interface
class CalendarService {
public:
    virtual void addEvent(string event) = 0;
};

// External API (incompatible)
class GoogleCalendarAPI {
public:
    void createEvent(string title) {
        cout << "Event added to Google Calendar: "
<< title << endl;
    }
};
```

```
// Adapter
class GoogleCalendarAdapter : public CalendarService { 16
private:
    GoogleCalendarAPI* googleAPI;

public:
    GoogleCalendarAdapter(GoogleCalendarAPI* api) {
        googleAPI = api;
    }

    void addEvent(string event) {
        googleAPI->createEvent(event);
    }
};

// Usage
int main() {
    GoogleCalendarAPI* googleAPI = new GoogleCalendarAPI();
    CalendarService* calendar = new
    GoogleCalendarAdapter(googleAPI);

    calendar->addEvent("Project Deadline");
}
```

Observer Pattern

Use Case: Notifications for Assignment Updates

```
#include <iostream>
#include <vector>
using namespace std;

// Observer
class Observer {
public:
    virtual void update(string message) = 0;
};

// Concrete Observer
class Student : public Observer {
    string name;
public:
    Student(string n) {
        name = n;
    }

    void update(string message) {
        cout << name << " received: " << message << endl;
    }
};

// Subject
class Assignment {
private:
    vector<Observer*> observers;
```

```
public:
    void addObserver(Observer* o) {
        observers.push_back(o);
    }

    void notifyObservers(string message) {
        for (auto o : observers) {
            o->update(message);
        }
    }

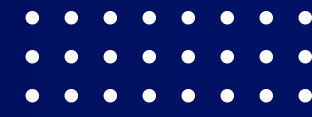
    void addAssignment(string task) {
        cout << "New Assignment: " << task << endl;
        notifyObservers("New assignment posted: " + task);
    }
};

// Usage
int main() {
    Assignment a;

    Observer* s1 = new Student("Ali");
    Observer* s2 = new Student("Sara");

    a.addObserver(s1);
    a.addObserver(s2);

    a.addAssignment("DBMS Project");
}
```



 StudySync

LoginGet Started

✦ Now with AI Study Planner

Sync Your Studies, Elevate Your Grades.

The all-in-one academic collaboration platform for university students. Manage assignments, collaborate with teams, and share resources seamlessly.

Start Syncing NowExplore Features



● 42 Team Members Online





Study Sync

Academic Collaboration Platform

[Login](#)[Register](#)

Email

 your.email@university.edu

Password

 Enter your password

[➔ Sign In](#)



Assignment Tracking

Never miss a deadline again



Team Collaboration

Work together seamlessly



Smart Scheduling

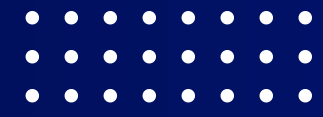
AI-optimized study plans



Resource Library

Share and discover materials

Register:



20



Study Sync

Academic Collaboration Platform

[Login](#) [Register](#)

Full Name

 Your full name

Email

 your.email@university.edu

Department

 e.g. Computer Science

Password

 Min 6 characters

 **Create Account**



Assignment Tracking

Never miss a deadline again



Team Collaboration

Work together seamlessly



Smart Scheduling

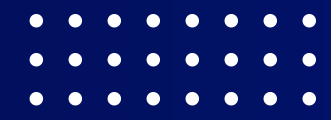
AI-optimized study plans



Resource Library

Share and discover materials

Forget Password:



21

A dark-themed user interface showing a 'Reset Password' dialog box. The dialog has a title bar with the text 'Reset Password' and a close button (an 'x' in a circle). Below the title bar, there is a text prompt: 'Enter your email address to reset your password.' followed by the label 'Email'. A text input field contains the email address 'your.email@university.edu' with a small envelope icon on the left. At the bottom of the dialog is a large, rounded rectangular button with the text 'Verify Email'.

Student Dashboard:



Study Sync

ACADEMIC PLATFORM

MAIN

 Dashboard

 Assignments

 Calendar

 Schedule

COLLABORATION

 Projects

 Resources

TOOLS

 AI Planner

 Notifications

 Simroz Schall

Student



Dashboard

Overview of your academic activities

Enrolled Courses

3

↑ Active this semester

Pending Assignments

0

✓ On track

Project Groups

0

👥 Active collaborations

Unread Notifications

0

✓ All caught up

Upcoming Deadlines

View All

Today's Schedule

Friday, May 1

Project Groups:





Study Sync

ACADEMIC PLATFORM

MAIN

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Simroz Sohail

Student



Project Collaboration

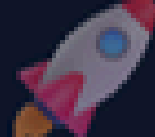
Collaborate with your team members

Search anything...



All GroupsActiveCompleted

+ Create Group

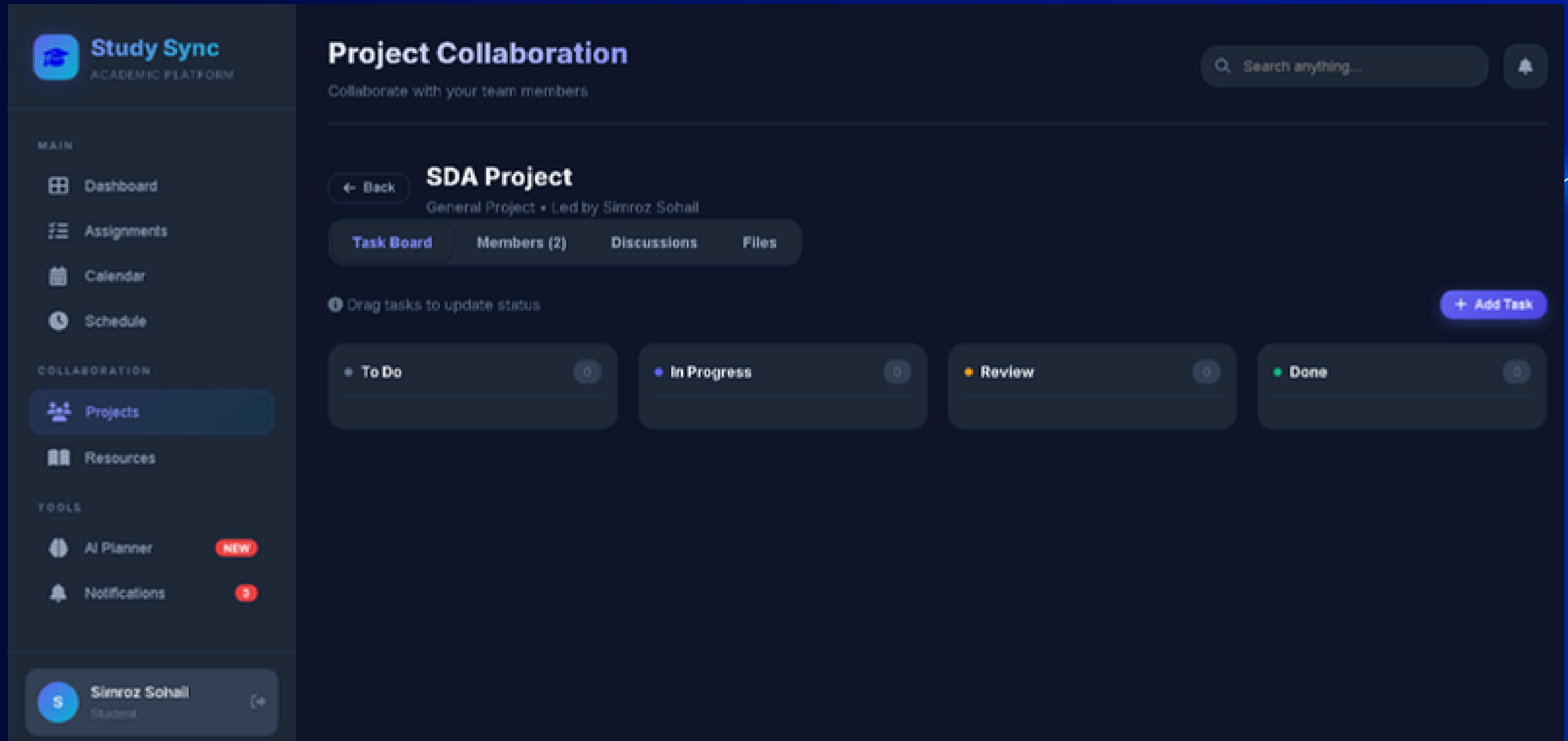


No project groups yet

Create a new group or join an existing one to start collaborating.

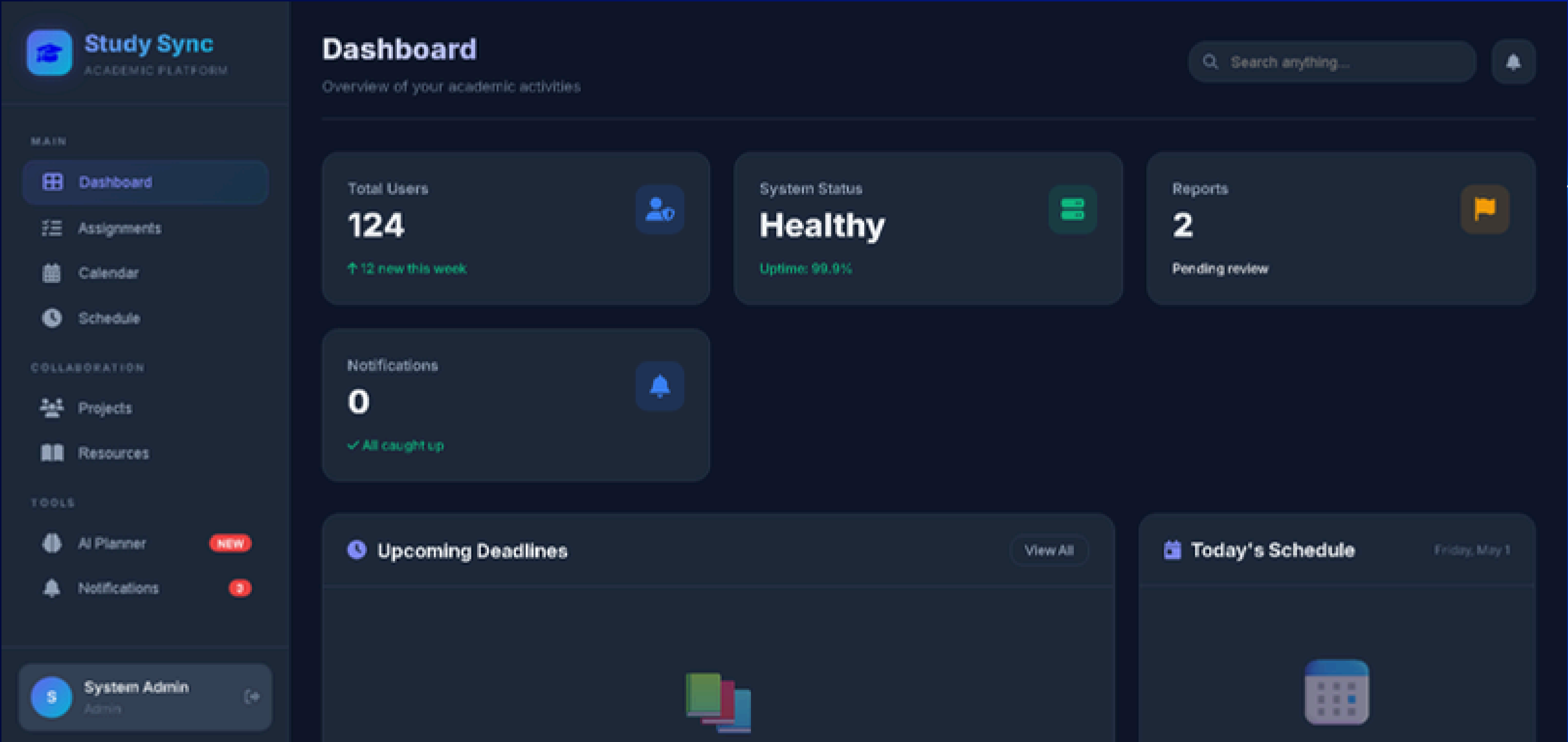
+ Create Your First Group

24

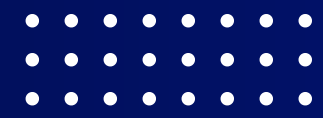




Admin Dashboard:



TA Dashboard:



27

**Study Sync**
ACADEMIC PLATFORM

MAIN

Dashboard

Assignments

Calendar

Schedule

COLLABORATION

Projects

Resources

TOOLS

AI Planner

Notifications

S

Sara Khan

+

Dashboard

Overview of your academic activities

My Courses

3

Active semester

Submissions

45

12 need grading

Avg. Grade

78%

Above semester avg

Notifications

0

All caught up

Upcoming Deadlines

View All

Today's Schedule

Friday, May 1

Instructor Dashboard:

28

**Study Sync**
ACADEMIC PLATFORM

MAIN

 Dashboard

 Assignments

 Calendar

 Schedule

COLLABORATION

 Projects

 Resources

TOOLS

 AI Planner NEW

 Notifications 3

 **Dr. Ahmed Qalser**
Instructor 

Dashboard

Overview of your academic activities

My Courses

3

Active semester



Submissions

45

12 need grading



Avg. Grade

78%

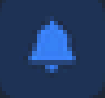
Above semester avg



Notifications

0

✓ All caught up



 Upcoming Deadlines

[View All](#)



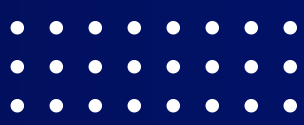
 Today's Schedule

Friday, May 1



29

Notifications:





Study Sync

ACADEMIC PLATFORM

MAIN

 Dashboard

 Assignments

 Calendar

 Schedule

COLLABORATION

 Projects

 Resources

TOOLS

 AI Planner

 Notifications



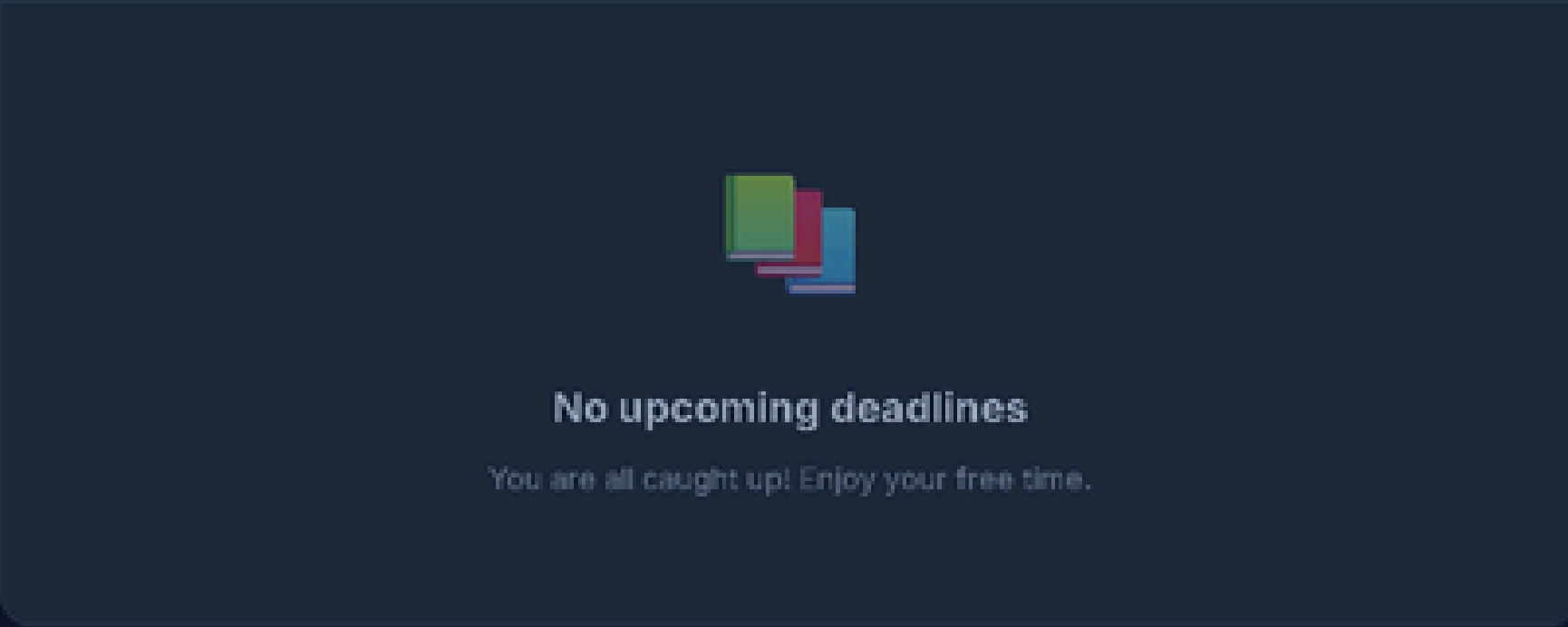
Simroz Sohail

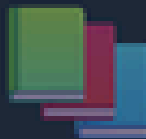
Student



 Upcoming Deadlines

View All





No upcoming deadlines

You are all caught up! Enjoy your free time.

 My Project Groups

View All



SDA Project

General Project

 1 members

 0 tasks

 Due: May 2, 2026

 Today's Schedule

Friday, May 1



No classes today

Enjoy your day off!

 Recent Resources



No resources yet

Thank
you